



Hipparchus observes the night sky



Ancient Chinese constellations

c.150 BCE

First great star catalogue

Hipparchus, a Greek astronomer, compiles the first great star catalogue of antiquity, grouping 850 stars into over 40 constellations. Hipparchus also divides the stars into six levels of brightness, the origin of the system of stellar magnitudes.

c.150

The *Almagest*

Greek astronomer Ptolemy produces a summary of Greek astronomy called the *Almagest*, which includes a revised version of Hipparchus's star catalogue with 48 constellations. It is the standard work on Western astronomy for nearly 1,500 years.

c.650

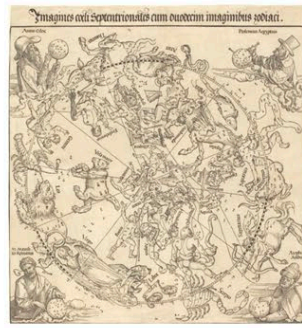
Oldest star chart

The oldest surviving star chart was drawn in 7th-century China on a paper scroll. Chinese constellations were smaller and more numerous than those in the West, with over 250 against Ptolemy's 48. Chinese astronomers also recorded hundreds more stars than the Greeks.

Tycho's observatory Uraniborg



The northern sky by Dürer



Taurus as depicted by al-Sufi



1598

Tycho Brahe

Tycho Brahe, a Danish astronomer, produces a new and improved catalogue of over 1,000 stars, ten times more accurate than the one in Ptolemy's *Almagest*. He still uses naked-eye sighting instruments, as the telescope has not yet been invented.

1515

Dürer's star chart

Albrecht Dürer draws the first European printed star chart, based on the catalogue in Ptolemy's *Almagest*. One half depicts the zodiac and northern constellations, the other shows the southern sky. Constellations are shown reversed, as on a celestial globe.

964

Arabic star charts

Al-Sufi, a Persian astronomer also known in the West as Azophi, produces an updated version of the Greek *Almagest*, entitled *The Book of the Fixed Stars*. This includes illustrations of each constellation, something the *Almagest* lacked, drawn in Arabic style.



The constellation Pegasus in *Uranographia*

1801

Greatest star atlas

The greatest of the old-style pictorial star atlases is published in 1801 by Johann Elert Bode, director of Berlin Observatory. Called *Uranographia*, it contains 17,000 stars divided into over 100 constellations, five of them invented by Bode himself.

1922–30

The final list

The newly formed International Astronomical Union (IAU) fixes the number of recognized constellations at 88, covering the entire celestial sphere, and draws up official boundaries for them. From now on, no more constellations can be added.

1989–93

Star cataloguing from space

A European Space Agency satellite called Hipparcos, named in memory of Hipparchus, compiles a catalog from orbit of the positions, motions, and brightnesses of over 100,000 stars with unprecedented accuracy.

2013

The Galaxy in 3D

The European observatory Gaia is launched. It will spend five years measuring the distances and motions of over a billion stars to build up a three-dimensional map of our Galaxy.

The Gaia spacecraft

